10. Fraction Operations

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Adding fractions with the same denominator:

Work out 5/9 + 2/9.

Answer: Add the numerators, keep the denominator the same: 5 + 2 = 7, so 5/9 + 2/9 = **7/9**.

Example 2 — Subtracting and simplifying:

Work out 9/10 − 4/10, and simplify if possible.

Answer: Subtract the numerators: 9 − 4 = 5, giving 5/10. This simplifies (divide top and bottom by 5): **1/2**.

Example 3 — Multiplying a fraction by a whole number:

Multiply 3/8 by 3, then write the result as a mixed number.

Answer: Multiply the numerator by the whole number, keep the denominator the same: 3 × 3 = 9, so 3/8 × 3 = 9/8. As a mixed number: **1 1/8**.

Example 4 — Finding a missing fraction:

3/9 + ? = 8/9. What fraction is missing?

Answer: Subtract to find the missing part: 8/9 − 3/9 = **5/9** (check: 3/9 + 5/9 = 8/9).

Example 5 — Multi-step word problem:

A ribbon is 11/12 m long. Priya cuts off 5/12 m, then her friend gives her 3/12 m more ribbon. How much ribbon does she have now?

Answer: First subtract: 11/12 − 5/12 = 6/12. Then add: 6/12 + 3/12 = 9/12, which simplifies to **3/4 m**.

Foundation

15 questions

1. 2/7 + 3/7

2. Which calculation shows 4/9 + 3/9? A) 7/9 B) 7/18 C) 12/9 D) 1/9

3. 7/8 − 2/8

4. Multiply 1/5 by 3.

5. What is 3/8 + 4/8? A) 7/8 B) 7/16 C) 12/8 D) 1/8

6. 6/7 − 2/7

7. A pizza is cut into 8 equal slices. Leo eats 3/8 and his sister eats 2/8. What fraction of the pizza have they eaten altogether?

8. Select ALL the additions below that give an answer of 6/10: A) 2/10 + 4/10 B) 1/10 + 5/10 C) 3/10 + 3/10 D) 4/10 + 1/10

9. 5/6 − 2/6

10. Multiply 3/7 by 2. A) 6/7 B) 6/14 C) 5/7 D) 3/14

11. 1/9 + 1/9 + 3/9

12. 8/11 − 3/11

13. Nadia has 4/10 of a cake and eats 3/10 more that was left over. What fraction has she eaten in total?

14. What is 1/8 + 1/8 + 1/8 + 1/8? A) 4/8 B) 1/8 C) 4/32 D) 1/32

15. 6/13 + 4/13

Standard

20 questions

1. 2/9 + 5/9

2. Work out 8/11 − 3/11. A) 5/11 B) 5/22 C) 11/11 D) 11/22

3. What is 3/10 + 3/10, simplified? A) 6/10 B) 3/5 C) 6/20 D) 3/20

4. Multiply 4/11 by 2.

5. 3/8 + ? = 7/8. What fraction is missing?

6. 9/10 − 4/10, simplified

7. A ribbon is 7/9 m long. Tom cuts off 3/9 m. How much ribbon is left?

8. Select ALL the subtractions below that give an answer of 2/5: A) 4/5 − 2/5 B) 3/5 − 1/5 C) 5/5 − 3/5 D) 4/5 − 1/5

9. 10/13 − 4/13

10. Multiply 3/8 by 2. A) 6/8 B) 6/16 C) 5/8 D) 3/16

11. 3/7 + 2/7, simplified

12. 4/15 + 5/15

13. A jug holds 10/12 litre of juice. Hassan pours out 6/12 litre. How much juice is left?

14. Which calculation is equal to 4/7 + 2/7? A) 6/14 B) 6/7 C) 2/7 D) 8/49

15. 9/14 − 3/14 + 2/14

16. 5/9 + 4/9, simplified if possible

17. Multiply 5/12 by 2.

18. 7/10 − ? = 2/10. What fraction is being subtracted?

19. A tank is 2/8 full. Rain adds another 3/8. What fraction of the tank is full now?

20. Select ALL the pairs of fractions below that add to exactly 1 whole: A) 3/10 and 7/10 B) 4/9 and 4/9 C) 5/12 and 7/12 D) 2/6 and 3/6

Hard

5 questions

1. What is 7/9 + 5/9, written as a mixed number in its simplest form? A) 1 1/3 B) 1 3/9 C) 12/9 D) 1 2/9

2. 4/7 + ? = 9/7. What fraction is missing?

3. Multiply 2/5 by 6, then write the answer as a fraction greater than 1 (an improper fraction).

4. A tank is 3/10 full. Rain adds another 4/10, then 2/10 leaks out. What fraction of the tank is full now?

5. Select ALL the pairs of fractions below that add to a result GREATER than 1 whole: A) 5/8 and 4/8 B) 3/9 and 4/9 C) 7/11 and 6/11 D) 4/7 and 2/7

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